Constrictive Pericarditis

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Adapted from: Constrictive Pericarditis

Listen to this Brick

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Differentiate the clinical presentation of constrictive pericarditis.
2. Explain the pathophysiology of constrictive pericarditis.
3. Interpret clinical findings to properly diagnose constrictive pericarditis.
4. Outline the management plan of constrictive pericarditis.

CASE CONNECTION

"It's getting harder to breathe, even when I talk. I've never been this sick, even when I had tuberculosis," says GZ, a 49-year-old man who presents to the clinic for evaluation. On exam, his jugular venous pressure (JVP) is elevated. Lower-extremity edema and findings consistent with ascites are detected. During inspiration, GZ's systolic blood pressure falls from 140 mm Hg to 124 mm Hg. During auscultation, an early diastolic sound that resembles a tapping noise is heard.

What is GZ's likely diagnosis? What diagnostic testing would be ordered to confirm the diagnosis? What would be the treatment options? Consider these questions as you read the brick, and we'll revisit GZ at the end.

GO TO CONCLUSION ↓

What Is Constrictive Pericarditis?

Constrictive pericarditis is a chronic inflammatory condition of the pericardium. Inflammation causes pericardial fibrosis that constricts the heart and impairs ventricular filling. Constrictive pericarditis is characterized by dense fibrous tissue, sometimes with calcification, that forms adhesions between the layers of the pericardium, making it inelastic or stiff. Constrictive pericarditis is a rare complication of acute pericarditis; however, it can be the end result of virtually any disease that involves the pericardium.

Constrictive pericarditis is chronic inflammation of the pericardium that results in pericardial fibrosis that constricts the heart and impairs ventricular filling.

How Do Patients With Constrictive Pericarditis Present? (LO #1)

Patients with constrictive pericarditis usually present with signs and symptoms relating to the impairment of the heart's diastolic filling; the impairment is caused by the heart's encasement by the inelastic pericardium. With impaired filling, cardiac output decreases, and patients may report dyspnea or fatigue on exertion. In addition, there might be signs of fluid overload, such as peripheral edema, ascites, and pleural effusions. This fluid overload occurs because blood returning to the heart cannot be accommodated by the stiff pericardium, leading to a backup of fluid into the systemic and pulmonary circulation.

The cause of the signs and symptoms seen in constrictive pericarditis is impaired diastolic filling of the heart (caused by the pericardium becoming inelastic).

What Is the Pathophysiology of Constrictive Pericarditis? (LO #2)

Causes

Constrictive pericarditis is a rare condition that can result from any pericardial disease, including acute pericarditis. In developed countries, you will find that constrictive pericarditis is often **idiopathic** (of unknown cause), but it can also result from viral infections, cardiac surgery, thoracic radiotherapy (primarily from Hodgkin disease or breast cancer), purulent pericarditis, or connective tissue disorders. However, in developing countries, **tuberculosis** is the leading cause of constrictive pericarditis and should always be considered in the differential diagnosis.

Consequences

The two main consequences of constrictive pericarditis are decreased cardiac output and decreased venous return, the latter of which causes venous congestion. These changes are primarily due to impairments in the ventricle filling process.

Decreased Cardiac Output. Pericardial fibrosis reduces the ease of filling (compliance) of the pericardium, preventing it from expanding as the heart receives incoming blood from the venous system.

As compliance decreases, the left ventricle end-diastolic volume also decreases. At this point, the pericardium starts exerting pressure on the heart (aka the “constriction”), causing elevation and equalization of the **end-diastolic pressures** in all four cardiac chambers. Because of this equalization, ventricular filling and cardiac output are greatly diminished, since small pressure differences between the chambers allow normal ventricular filling during diastole.

Another cause of decreased cardiac output is the enhancement of **ventricular interdependence**. In simple terms, this is the way that a change in the volume of one ventricle will alter the volume of the other ventricle. Normally, with increases in venous return (e.g., during inspiration when negative thoracic pressure develops), the volume of blood in the right ventricle (RV) causes the interventricular septum to bulge into the left ventricle, reducing the LV volume.

Normally, the LV size decreases only slightly. However, in patients with constrictive pericarditis, where the pericardium is fibrotic and inelastic, increases in right ventricular volume cannot be well accommodated, so the interventricular septum shifts toward the left, markedly decreasing the left ventricular filling. This results in decreased stroke volume and cardiac output during inspiration (Figure 1).

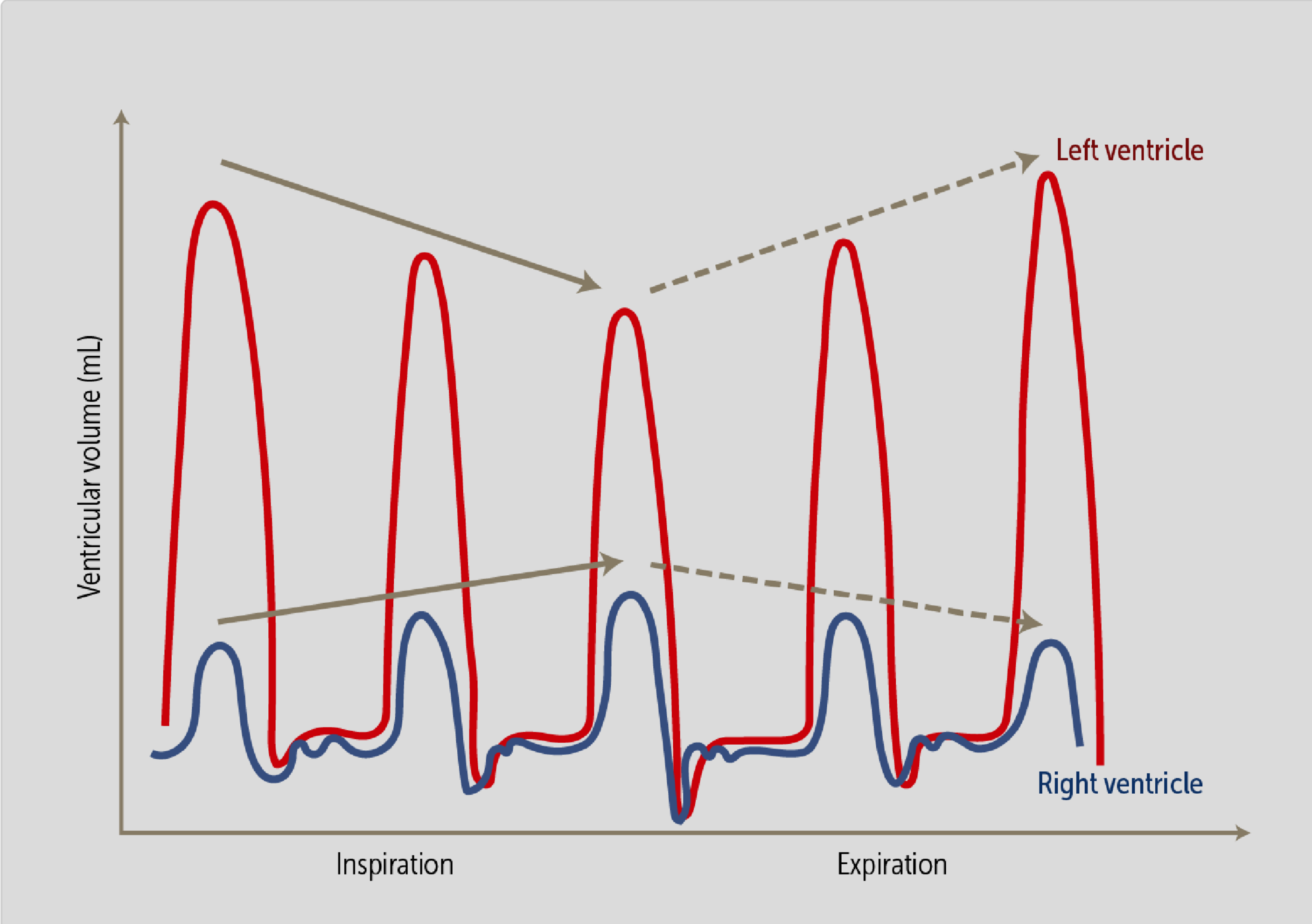

Figure 1. Increased Ventricular Interdependence in Constrictive Pericarditis.

The two main hemodynamic changes that lead to decreased cardiac output are equalization of end-diastolic pressures in all four cardiac chambers and enhanced ventricular interdependence.

Decreased Venous Return. A second consequence of constrictive pericarditis is a decrease in venous return. Recall that inspiration normally causes a negative intrathoracic pressure, which is transmitted to the large thoracic veins and the heart (right atrium), causing their pressure to fall. This increases the pressure gradient between the extrathoracic veins and the right atrium, effectively creating a "thoracic pump" that pulls blood back into the heart.

However, in constrictive pericarditis, this lower intrathoracic pressure does not transmit to the heart because of the fibrosed pericardium. Venous return drops, causing venous congestion (e.g., JV distention). The decreased venous return will lower preload and also contribute to reduced cardiac output.

How Do We Diagnose Constrictive Pericarditis? (LO #3)

The diagnosis of constrictive pericarditis is challenging because its signs and symptoms are often nonspecific and can be falsely attributed to other causes. Hence, diagnosis requires a high degree of clinical suspicion. The findings from the patient's history and physical exam combined with those from echocardiography are usually sufficient to make the diagnosis.

History

As mentioned earlier, constrictive pericarditis can present with signs and symptoms of reduced cardiac output and fluid overload. The presence of these findings helps direct the rest of the diagnostic evaluation.

Physical Examination

Jugular venous pressure (**JVP**) is elevated in patients with constrictive pericarditis, resulting in jugular venous distention (JVD). A key feature that separates constrictive pericarditis from other causes of elevated JVP is the presence of a **prominent y descent** on JVP waveforms, which reflects a more rapid than usual emptying of the right atrium. (The JVP waveform is a visual representation of the changing pressure within the right atrium during the cardiac cycle, observed indirectly through pulsation in the internal jugular vein.)

Another finding on examination of the jugular veins is **Kussmaul sign**, defined as a lack of the normal JVP decrease during inspiration. Normally, inspiration lowers the intrathoracic pressure, increasing the venous return to the right atrium and subsequently decreasing the JVP. But because of the limited pericardial expansion in constrictive pericarditis, the JVP does not decrease with inspiration—on the contrary, it often increases, causing JVD. This is a nonspecific finding, and we can see this finding in other disorders such as right-sided heart failure or tricuspid regurgitation.

Pulsus paradoxus is defined as a decrease in systolic blood pressure by more than 10 mm Hg during inspiration (Figure 2). This occurs in constrictive pericarditis because the enhanced ventricular interdependence lowers cardiac output, as described above. However, this finding is not specific to constrictive pericarditis and can be seen in other disorders such as cardiac tamponade and some respiratory diseases.

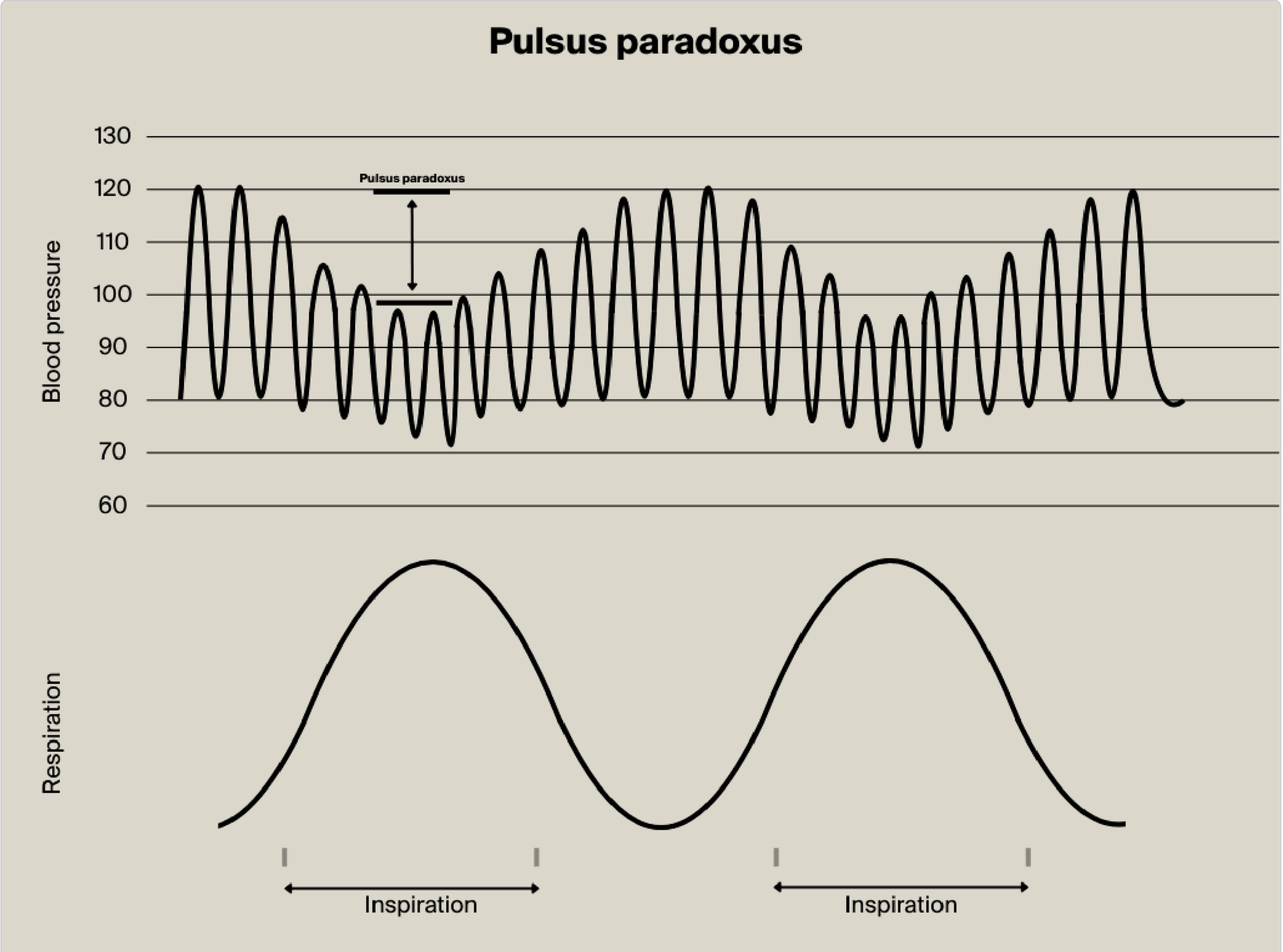

Figure 2. Pulsus Paradoxus. Note the drop in systolic blood pressure associated with inspiration.

A **pericardial knock** is a high-pitched early diastolic sound, corresponding to the abrupt cessation of ventricular filling as the ventricles stop expanding once the pericardial elastic limit is reached.

CLINICAL CORRELATION

Here is a link to hear a pericardial knock:

https://www.youtube.com/watch?v=5_EkVuMeNRA&t=24s

Electrocardiography

An ECG generally does not reveal any uniquely identifying features, and it may even be normal. Nonspecific changes may include low QRS voltages and generalized T-wave flattening or inversion. Some patients may have atrial fibrillation.

Radiology Studies

A chest x-ray is usually normal, but it may demonstrate pericardial calcium deposits in a minority of patients (Figure 3).

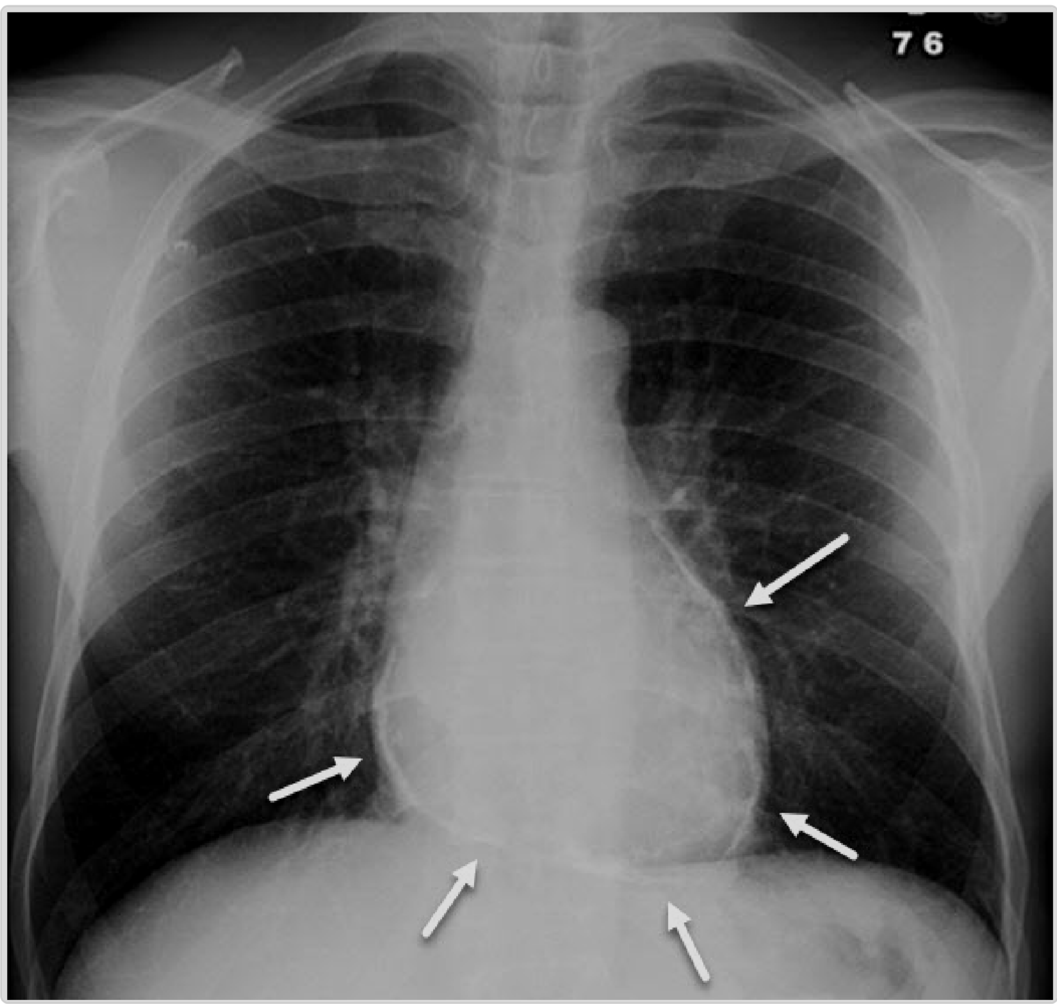

Figure 3. Calcifications in Constrictive Pericarditis. Note the curvilinear calcifications (white arrows) surrounding the heart in this patient with constrictive pericarditis.

Echocardiography is usually sufficient to confirm the diagnosis of constrictive pericarditis. It reveals the extent of pericardial fibrosis, as demonstrated by many parameters such as increased pericardial thickness or abnormal bulging of interventricular septum with inspiration. Sometimes, impaired diastolic filling in a constrictive pattern can be recognized.

Computed tomography (CT) or magnetic resonance imaging (MRI) may be used as alternatives to echocardiography in uncertain cases or when detailed anatomical visualization is required for surgical planning.

Invasive Hemodynamic Studies

In cases in which constrictive pericarditis is highly suspected, the gold standard diagnosis is invasive hemodynamic evaluation through cardiac catheterization. By using pressure catheters inside the right and left ventricles, the hemodynamic changes described above can be observed, such as elevation and equalization of diastolic blood pressures and exaggerated ventricular interdependence. There may also be elevated right atrial pressure or prominent y descent in the jugular venous tracing.

Echocardiography is most useful for diagnosing constrictive pericarditis.

How Do We Manage Constrictive Pericarditis? (LO #4)

Management is often conservative, with medications alone. Surgery may be needed in severe cases.

Medications

Fibrotic tissue does not respond well to anti-inflammatory drugs, but if there is ongoing acute pericarditis, nonsteroidal anti-inflammatory drugs may be useful. Diuretics may be used to improve pulmonary and systemic congestion in patients with constrictive pericarditis, but these agents must be used with caution. Because the heart is compressed by fibrotic tissue and has low cardiac output, any drop in intravascular volume can lead to a further lowering of cardiac output, leading to hypotension.

Surgery

The only definitive treatment for patients late in the disease process is **pericardiectomy**, which is the removal of a portion or all of the pericardium (Figure 4). It is used for markedly symptomatic patients and often improves symptoms, but the operative risk is high (4%-12% operative mortality), so patients should be carefully selected.

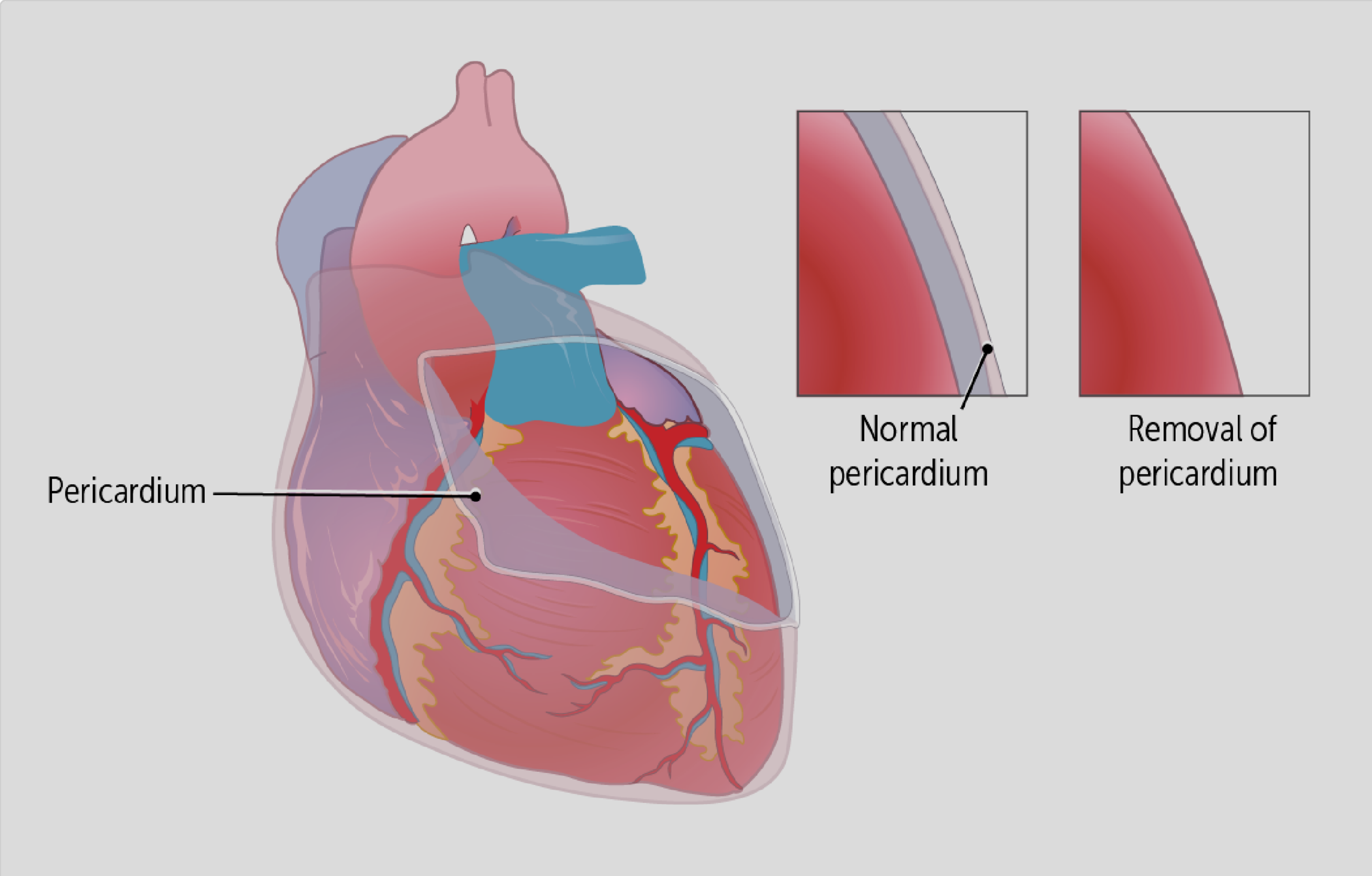

Figure 4. Pericardiectomy (Surgical Removal of the Pericardium).

Pericardiectomy is the name of the surgical procedure used to treat constrictive pericarditis.

CASE CONNECTION

BACK TO INTRODUCTION

Thinking back to GZ, what is the likely diagnosis? What diagnostic testing is ordered to confirm the diagnosis? What are the treatment options?

Constrictive pericarditis is the suspected diagnosis. GZ's thickened and fibrotic pericardium leads to impaired diastolic filling, and this causes his presenting signs and symptoms. As GZ learns this, he asks, "What caused this to happen?" You tell him that this could be due to his prior history of tuberculosis and resultant chronic inflammation of the pericardium. An echocardiogram is ordered, and the possible need for a cardiac catheterization is discussed with GZ. Medical therapy is started with diuretics. The likely need for surgical treatment, a pericardiectomy, is also discussed with GZ. "If it will help me breathe better, let's go for it," GZ says. "I need to feel better and get back to work."

Summary

- Constrictive pericarditis is a form of chronic inflammation involving formation of fibrous tissue throughout the pericardium.
- Constrictive pericarditis happens when the amount of blood allowed to fill the heart chambers is restricted by the fibrous tissue of the thickened pericardium.
- Patients present with signs and symptoms of reduced cardiac output, such as dyspnea on exertion, or fluid overload, such as peripheral edema.
- Constrictive pericarditis can be the end result of any pericardial disease.

- Constrictive pericarditis reduces cardiac output by impairing normal ventricular filling during diastole, equalizing end-diastolic pressures in all four cardiac chambers, and enhancing ventricular interdependence; thus, LV volume during inspiration is reduced.
- Physical exam findings in constrictive pericarditis include elevated jugular venous pressure, pulsus paradoxus, Kussmaul sign, and pericardial knock.
- ECG reveals nonspecific changes in constrictive pericarditis.
- Chest x-ray may reveal pericardial calcium deposits in constrictive pericarditis.
- The diagnostic confirmation of constrictive pericarditis is usually made through echocardiography, although CT or MRI are reasonable alternatives.
- Invasive hemodynamic evaluation is performed to confirm the diagnosis.
- Medical management for constrictive pericarditis includes anti-inflammatory drugs and diuretics.
- Pericardiectomy, a surgical treatment, may improve symptoms, but the operative risk is high.

Review Questions

1. Which of the following is least associated with constrictive pericarditis?

A. Jugular venous distention

B. Pericardial calcium deposits on chest x-ray

C. Pericardial knock on cardiac auscultation

D. Pulsus paradoxus

E. Reduced pericardial thickness on echocardiogram

Explanation (requires correct answer)

2. Which of the following is the most common cause of constrictive pericarditis in developing countries?

A. Connective tissue disorders

B. Postcardiac surgery

C. Postradiation therapy

D. Tuberculosis

E. Viral infections

Explanation (requires correct answer)

3. A 65-year-old male slowly develops constrictive pericarditis over several years after receiving thoracic radiation for a malignancy. He now reports increasing fatigue, dizziness, and dyspnea, and he has bilateral pleural effusions and peripheral edema on exam. Which of the following is the best next step in management?

A. Colchicine

B. Furosemide

C. Ibuprofen

D. Pericardiectomy

E. Prednisone

Explanation (requires correct answer)

How would you rate this Brick?

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 **Differentiate the clinical presentation of constrictive pericarditis.**

02 **Explain the pathophysiology of constrictive pericarditis.**

03 **Interpret clinical findings to properly diagnose constrictive pericarditis.**

04 **Outline the management plan of constrictive pericarditis.**

Objective Confidence: 0/4